SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: Theory of Computation **COURSE CODE: CSA13**

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems(BL:3)

Assessment Tool Weightage: 10%

QUESTIONS: 4 Total Marks:20 Duration : 1 Hour

Technical Presentation

Code of Conduct:

I J. SAI DHEERAJ (192311191) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

action. Signature of the student: J. SAI DHEERAJ

SIMATS ENGINEERING

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Saveetha Institute of Medical and Technical Sciences
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Smart ATM Transaction Validation System Using State Machine

PRESENTED BY

Y Joy Winston (192411158)

K Arun Kumar (192372287)

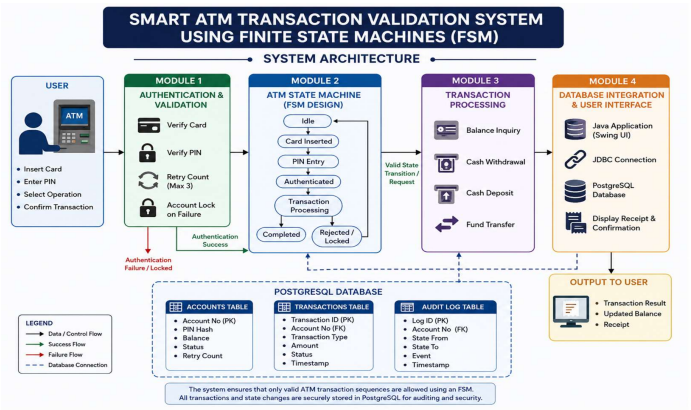
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GUIDED BY
DR. K. Baskar

Abstract

- ATMs are used for important banking transactions, so they need strong security.
- This project uses a Finite State Machine (FSM) to control the correct order of ATM operations.
- It allows actions like card insertion, PIN entry, transaction selection, and confirmation only in the correct order.
- If the user performs an invalid or unauthorized action, the system rejects it.
- Java is used to develop the ATM system, and PostgreSQL is used to store account, transaction, and audit information.
- The system helps make ATM transactions more secure, reliable, and easy to monitor.





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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (8 Marks)	Level 3 – Proficient (6 Marks)	Level 2 – Developing (4 Marks)	Level 1 – Beginning (2 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application

Completeness of Response	**15**	**Covers all required parts of the question**	**Covers most parts with minor omissions**	**Covers only some parts**	**Incomplete or missing response**
Clarity & Presentation	**10**	**Clear, concise, and well organized answer**	**Understandable with minor issues**	**Somewhat unclear or poorly structured**	**Difficult to understand**